



DEPARTMENT OF  
**DESIGN AND  
MERCHANDISING**



## Biomimicry, Industrial Ecology & Biophilic Design

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# Question

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How do electric cars communicate?



# Question

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Last I checked cars do not  
talk



# Question

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If they did, they would say  
'watts up'



## World Population

<https://www.worldometers.info/world-population/>

Current World Population  
**8,313,808,370**

[view all people on 1 page >](#)

### TODAY

Births today

**117,318**

 Deaths today

**55,281**

Population Growth today

**62,037**

### THIS YEAR

Births this year

**87,889,891**

Deaths this year

**41,414,129**

Population Growth this year

**46,475,762**



# Native American Perspective

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# Sustainability

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- Let's take a step back
  - Bruntland Report (1987):
    - “Development which meets the needs of the present without prohibiting future generations to meet their own needs.”
  - Interrelationship of environment, society, and economic systems
  - Intergenerational Responsibility



# Toolbox

Conscious emulation of life's genius

~Janine Benyus

<https://toolbox.biomimicry.org/introduction/>



## Biomimicry

*Making nature an important resource for problem solving*

[https://www.bbc.com/news/av/science-environment-48152165#google\\_vignette](https://www.bbc.com/news/av/science-environment-48152165#google_vignette)

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We are not alone





# TED Talk

Janine Benyus

- [http://www.ted.com/talks/janine\\_benyus\\_biomimicry\\_in\\_action?language=en](http://www.ted.com/talks/janine_benyus_biomimicry_in_action?language=en)



# Costa Rica

<https://youtu.be/eAJHTWT5o50>



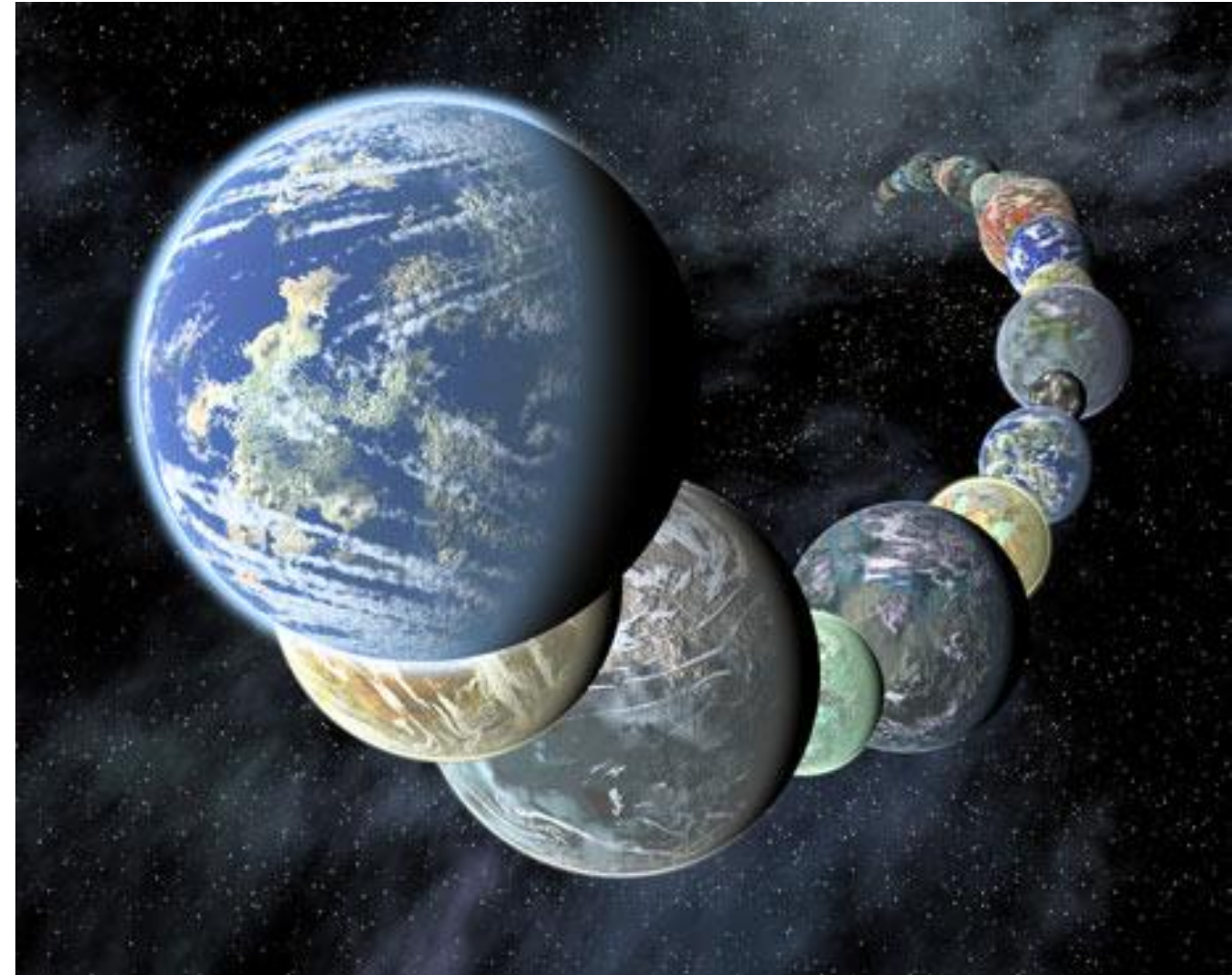
*How studying*  
**BIRD WING**  
*inspired ...*



Inspiration

<https://www.learnbiomimicry.com/blog/best-biomimicry-examples>

# Life's tenets

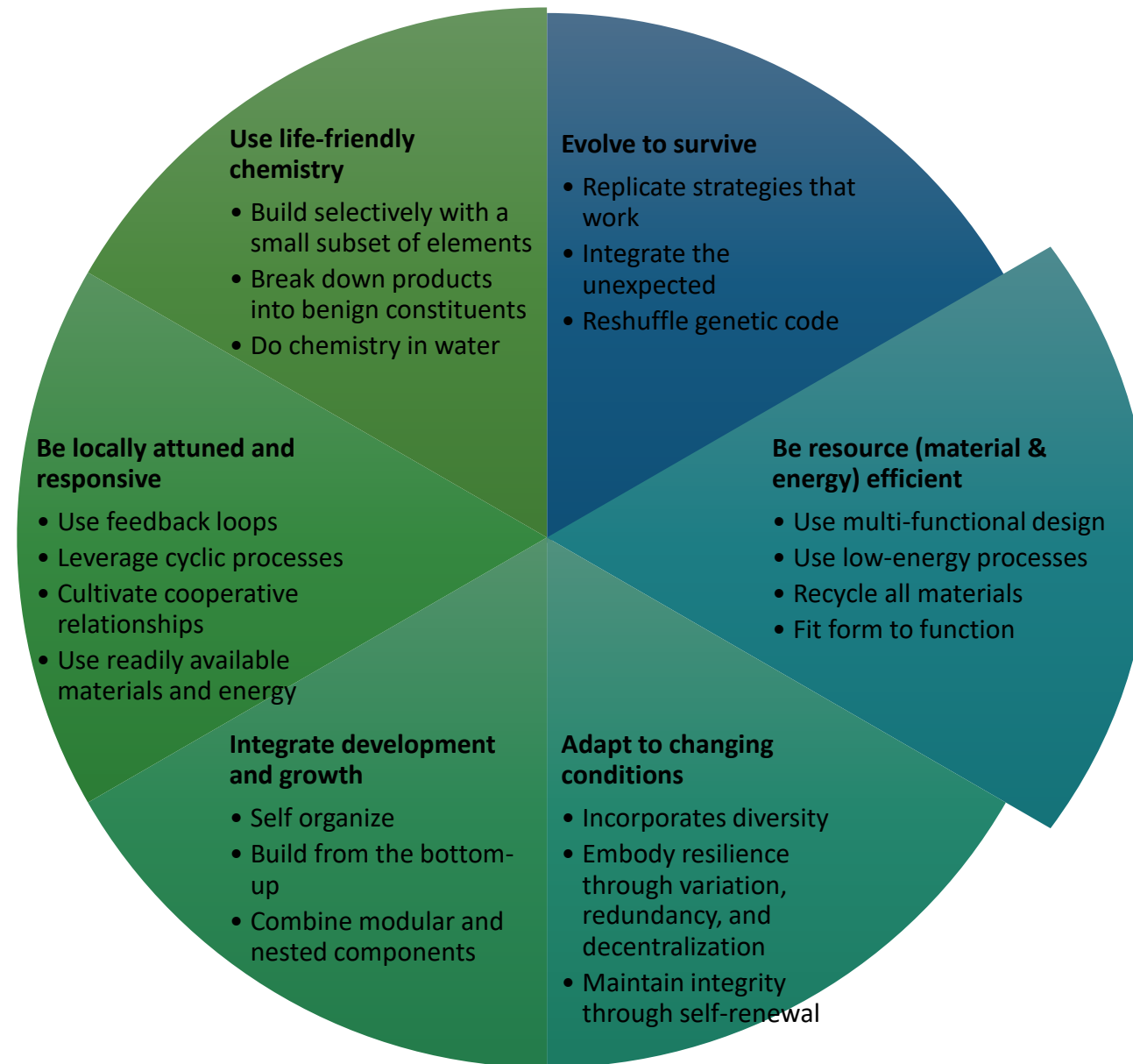


Life creates conditions conducive to life  
Life adapts & evolves

# Life's Principles

## Design Lessons from Nature

In NATURE...



# Biomimicry

- What is it?
  - **Biomimicry** is an approach to innovation that seeks sustainable solutions to human challenges by emulating nature's time-tested patterns and strategies. The goal is to create products, processes, and policies—new ways of living—that are well-adapted to life on earth over the long haul



# Biomimicry: Levels

- Emulating nature's structure and/or infrastructure: FORM
- Biomimicry + biology (using biological processes to design): PROCESS
- Biomimicry + synthetic biology\* (reengineering with biological processes): PROCESS
- Biomimicry + natural ecosystem: SYSTEM (industrial ecology)

\*Most controversial

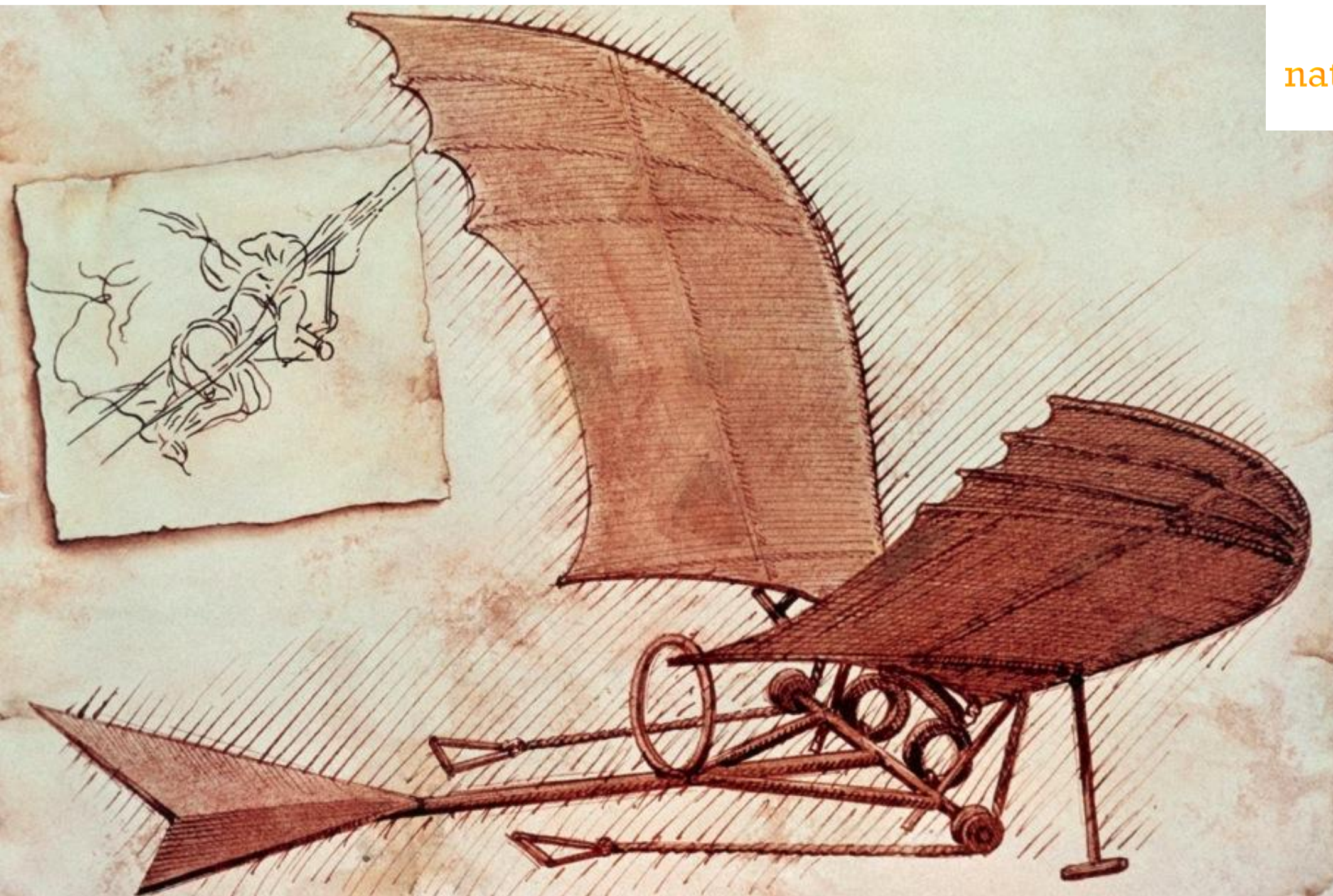




Struts & Beams

Mimicking nature's structure  
and infrastructure

Mimicking  
nature's structure  
and  
infrastructure

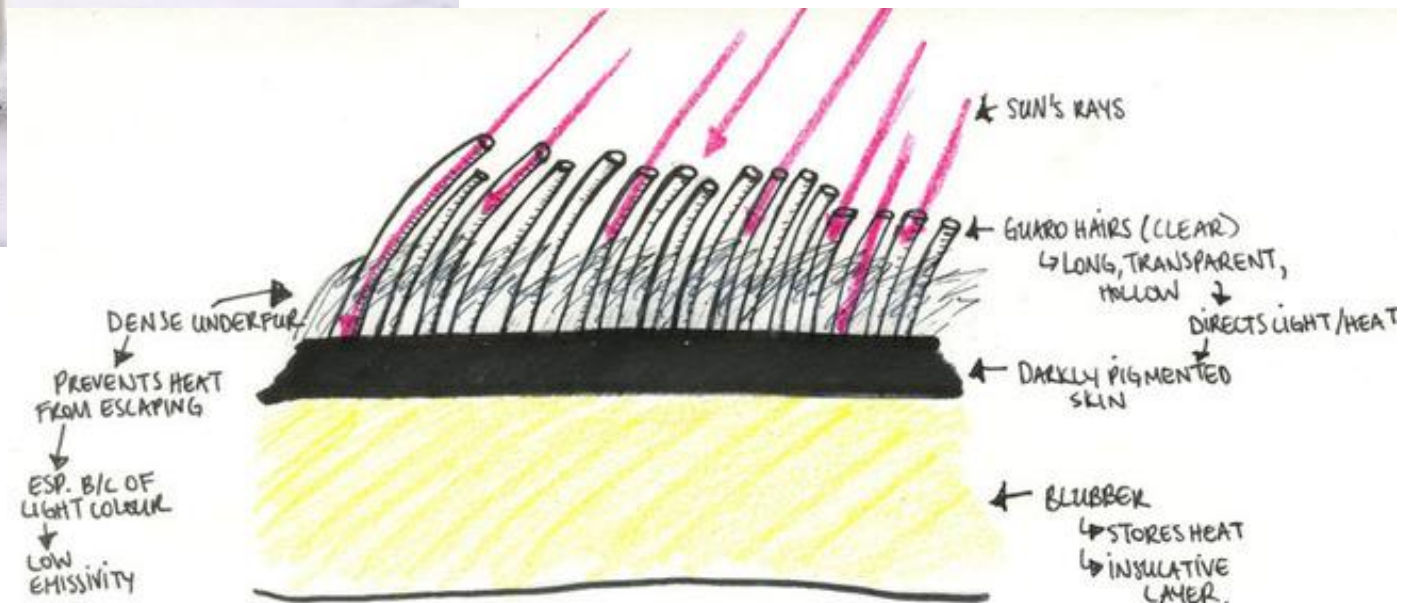


Leonardo da  
Vinci's flying  
machine

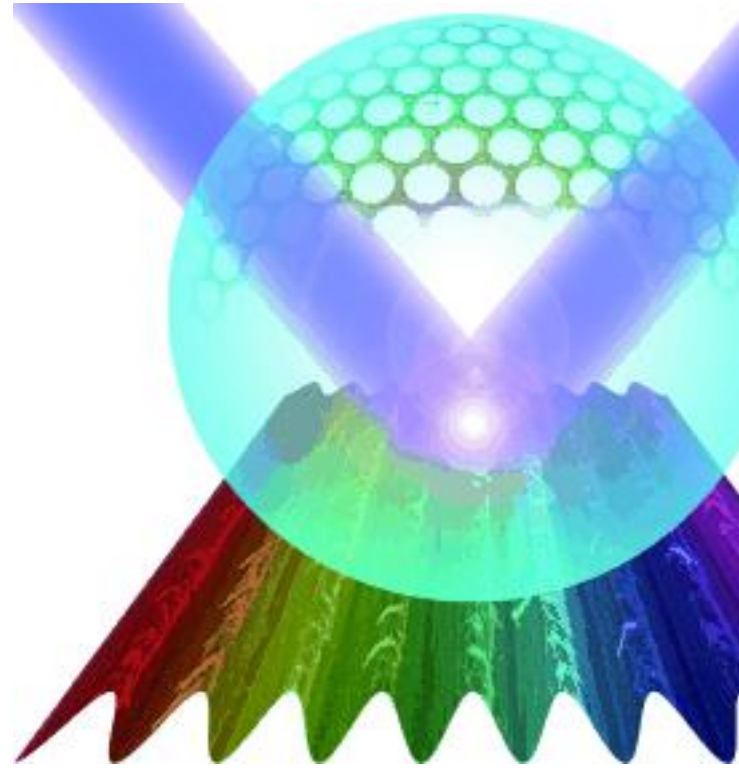
# Mimicking nature's structure and infrastructure



## Structural Color



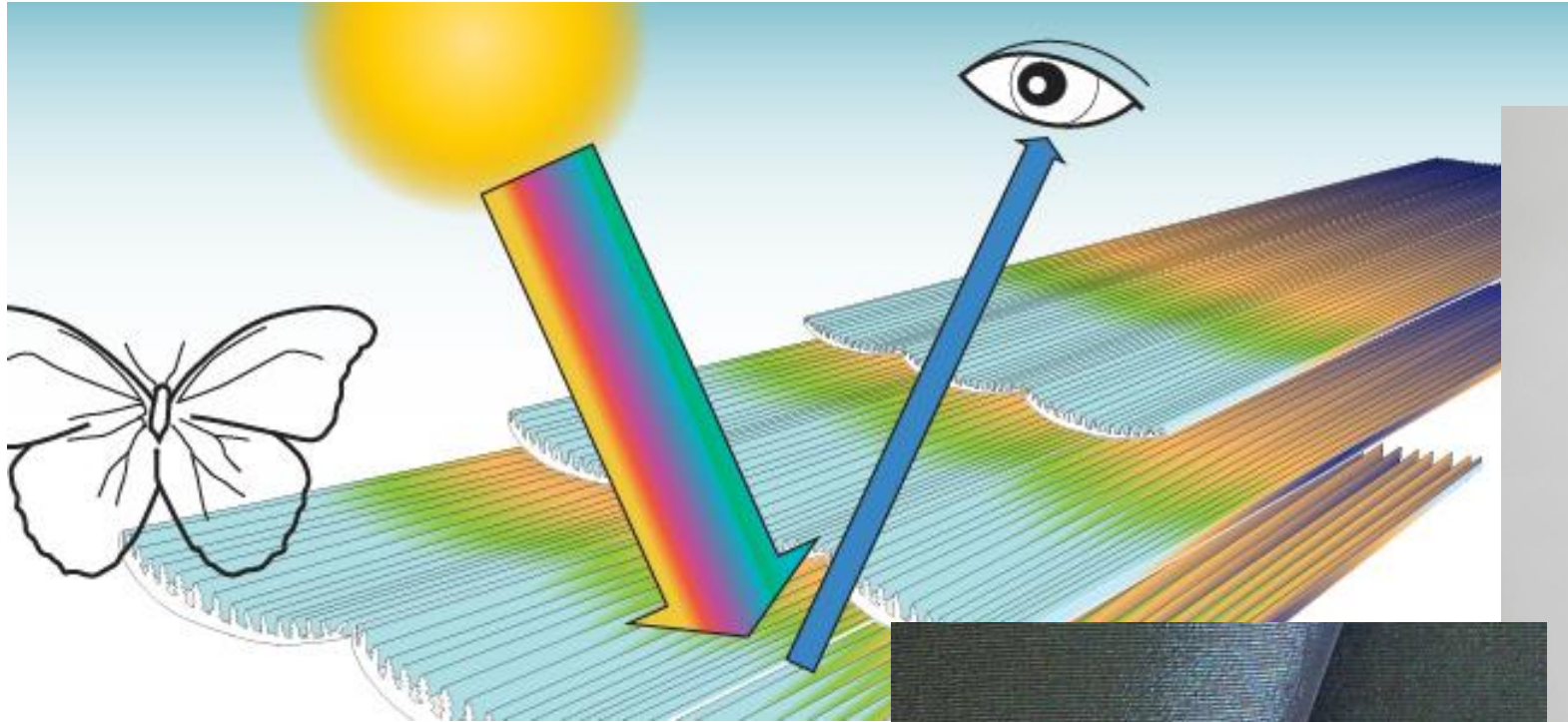
## Mimicking nature's structure and infrastructure



## Structural Color



# Mimicking nature's structure and infrastructure



Structural Color

Biomimicry +  
biology (using  
biological process  
to design)



Solar  
panels

Suzanne Lee

Growing textiles for a  
“leather jacket”



Biomimicry + biology  
(using biological process to  
design)

Biomimicry + biology  
(using biological process to  
design)



Carole Collett's hydroponic  
greenhouse at Textile Futures  
Research Center: University of  
the Arts London (Central St.  
Martins)

Using nature's models to grow  
textiles for applications to  
clothing & wallpaper



## Catalytic Textiles



Biomimicry + biology  
(using biological process to  
design)



[http://www.ted.com/talks/nina\\_tandon\\_caring\\_for\\_cells](http://www.ted.com/talks/nina_tandon_caring_for_cells)



Biomimicry + Synthetic Biology  
Controversial

# Mitchell Joachim

- [https://en.wikipedia.org/wiki/Fab\\_Tree\\_Hab](https://en.wikipedia.org/wiki/Fab_Tree_Hab)
- [https://www.ted.com/talks/mitchell\\_joachim\\_don\\_t\\_build\\_your\\_home\\_grow\\_it?language=en](https://www.ted.com/talks/mitchell_joachim_don_t_build_your_home_grow_it?language=en)



Grow your House

- Neri Oxman

- [https://www.ted.com/talks/neri\\_oxman\\_design\\_at\\_the\\_intersection\\_of\\_tech\\_nology\\_and\\_biology#t-561813](https://www.ted.com/talks/neri_oxman_design_at_the_intersection_of_tech_nology_and_biology#t-561813)
- [https://en.wikipedia.org/wiki/Neri\\_Oxman](https://en.wikipedia.org/wiki/Neri_Oxman)



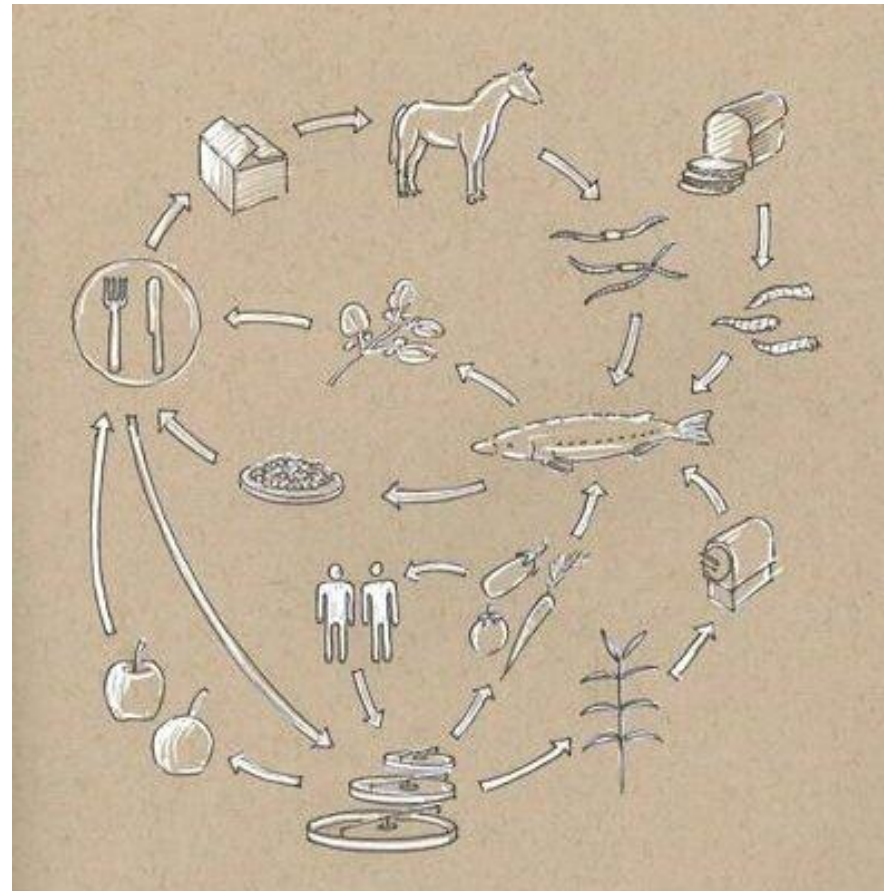
Technology & Biology

**Industrial ecology** (IE) is the study of material and energy flows through **industrial** systems. The global **industrial** economy can be modeled as a network of **industrial** processes that extract resources from the Earth and transform those resources into commodities which can be bought and sold to meet the needs of humanity



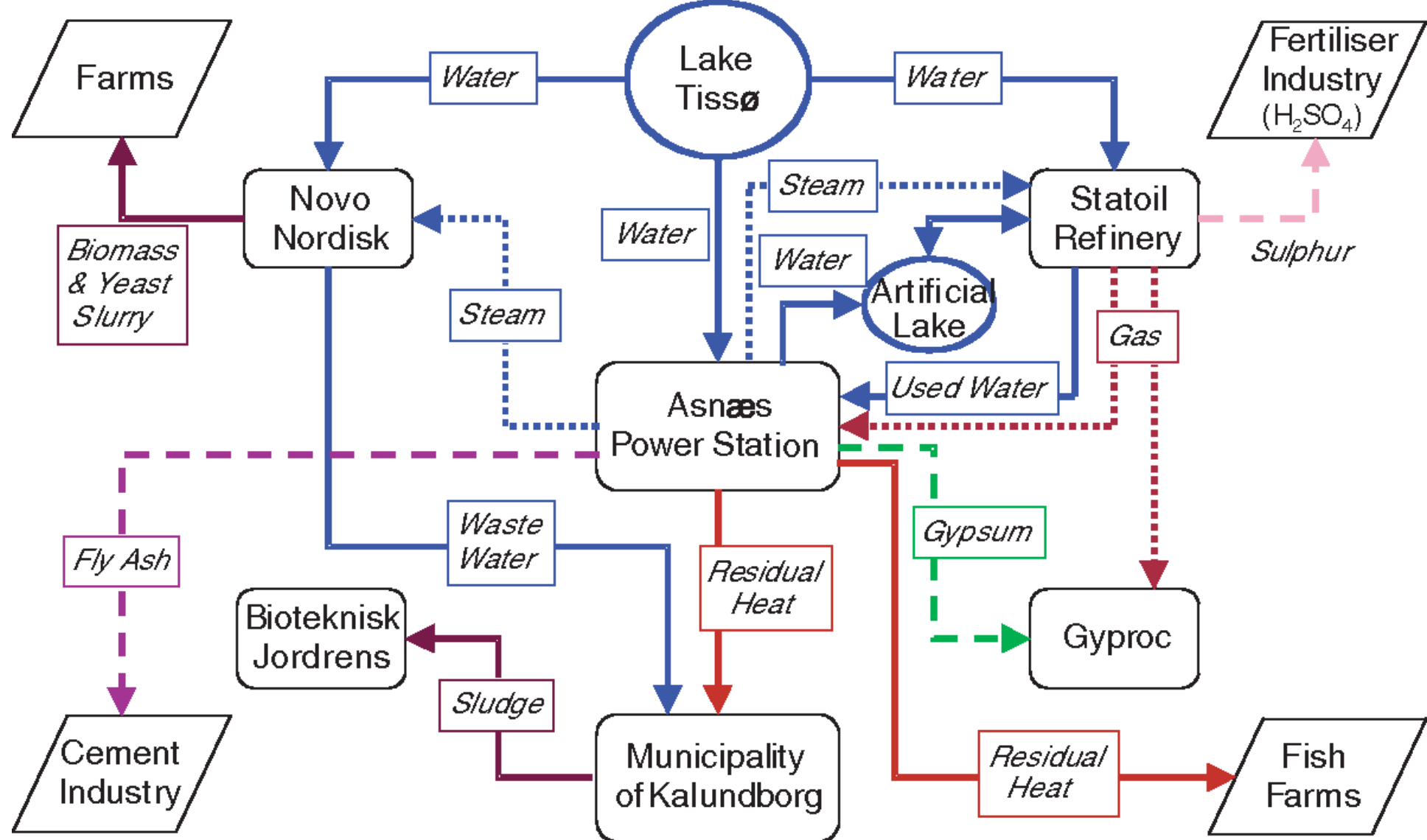
Michael Pawlyn

[https://www.ted.com/talks/michael\\_pawlyn\\_using\\_nature\\_s\\_genius\\_in\\_architecture/up-next](https://www.ted.com/talks/michael_pawlyn_using_nature_s_genius_in_architecture/up-next)



1. The restaurant paid Wiles to remove their cardboard boxes, which he would then shred.
2. Wiles sold the cardboard as bedding to a horse stable.
3. The bedding was then collected and used in vermicompost (worms) production.
4. The worms were then used to feed Sturgeon, which produce caviar (the more expensive stage of the process).
5. The caviar was then sold back to the original restaurant that sold him the cardboard, effectively closing the loop.

IE – Example 1



## IE – Example 2

- Love of life or living systems
- Urge to affiliate with other forms of life
- “Tendency to focus on life and life-like processes has a genetic basis”

~ Edward O Wilson (1984)

- “Humans intrinsically like and find pleasurable environments that contain key features of the savannah that were most likely to have aided our ancestors’ survival and well being”

~ Savannah Hypothesis

- “Connections we subconsciously seek with the rest of life”
- Compare to phobia
- Adult mammals
  - Baby mammal faces
- Biophilia hypothesis
  - Instinctive bond  $\leftrightarrow$  Humans & Living systems
- Our love for natural life
  - Helps sustain life!!
  - Benyus?



# Biophilia

- <http://www.fallingwater.org/>
- Frank Lloyd Wright
  - 1936-1939
- Pennsylvania



Fallingwater



Thorncrown Chapel

[https://en.wikipedia.org/wiki/Thorncrown\\_Chapel](https://en.wikipedia.org/wiki/Thorncrown_Chapel)

## NATURE IN THE SPACE



## NATURAL ANALOGUES



## NATURE OF THE SPACE



### 1. Visual Connection with Nature

A view to elements of nature, living systems and natural processes.

### 2. Non-Visual Connection with Nature

Auditory, haptic, olfactory, or gustatory stimuli that engender a deliberate and positive reference to nature, living systems or natural processes.

### 3. Non-Rhythmic Sensory Stimuli

Stochastic and ephemeral connections with nature that may be analyzed statistically but may not be predicted precisely.

### 4. Thermal & Airflow Variability

Subtle changes in air temperature, relative humidity, airflow across the skin, and surface temperatures that mimic natural environments.

### 5. Presence of Water

A condition that enhances the experience of a place through the seeing, hearing or touching of water.

### 6. Dynamic & Diffuse Light

Leveraging varying intensities of light and shadow that change over time to create conditions that occur in nature.

### 7. Connection with Natural Systems

Awareness of natural processes, especially seasonal and temporal changes characteristic of a healthy ecosystem.

### 8. Biomorphic Forms & Patterns

Symbolic references to contoured, patterned, textured or numerical arrangements that persist in nature.

### 9. Material Connection with Nature

Material and elements from nature that, through minimal processing, reflect the local ecology or geology to create a distinct sense of place.

### 10. Complexity & Order

Rich sensory information that adheres to a spatial hierarchy similar to those encountered in nature.

### 11. Prospect

An unimpeded view over a distance for surveillance and planning.

### 12. Refuge

A place for withdrawal, from environmental conditions or the main flow of activity, in which the individual is protected from behind and overhead.

### 13. Mystery

The promise of more information achieved through partially obscured views or other sensory devices that entice the individual to travel deeper into the environment.

### 14. Risk/Peril

An identifiable threat coupled with a reliable safeguard.

# Birch tree and moss garden in the New York Times Building in New York City



## 1. Visual connection with nature

# Fountain & garden in the Calat Alhambra in Granada, Spain



## 2. Non-visual connection with nature

# Dockside Green Community, Vancouver



## 3. Non-rhythmic sensory stimuli

## 4. Thermal and airflow variability



# Robert & Arlene Kogod Courtyard Smithsonian American Art Museum, DC



## 5. Presence of Water



# Greenroof of COOKFOX architects, NY



## 7. Connection with Natural Systems

# Façade of Manuel Ge Gonzalez Hospital, Mexico



## 8. Biomorphic forms and patterns



## 9. Material Connection with Nature

# Summer Palace, Beijing, China



## 10. Complexity & Order

# Central Plaza of Salk Institute, CA

Louis Kahn in La Jolla, California frames the view of the Pacific



11. Prospect

# Protected seating alcoves along the Henderson Bridge, Singapore



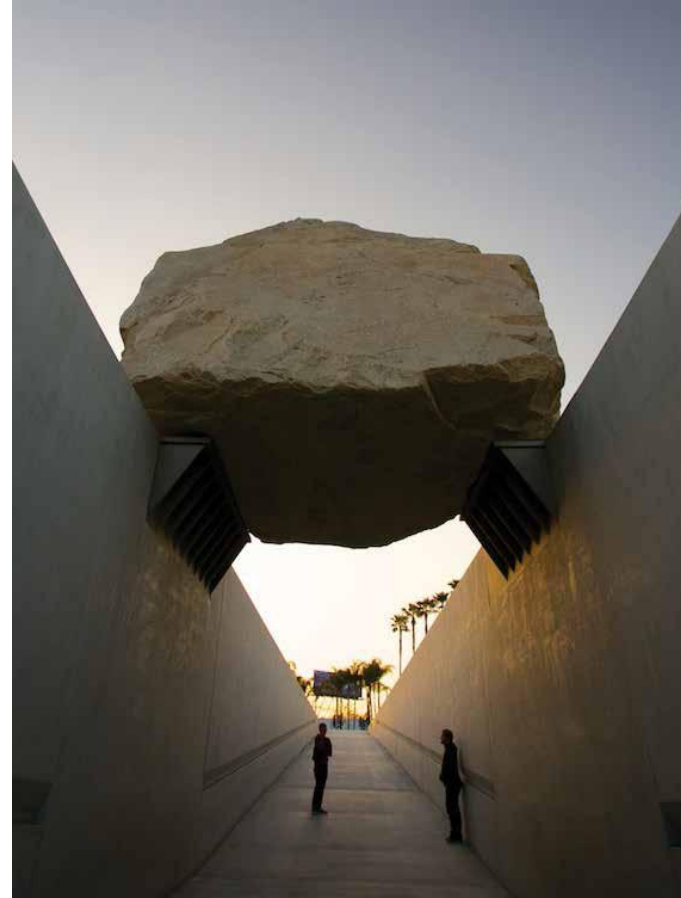
## 12. Refuge

# Lan Su Chinese Garden, Portland, OR



## 13. Mystery

# The Levitated Mass at Los Angeles County Museum of Art



14. Risk/Peril

| 14 PATTERNS         | *                                 | STRESS REDUCTION                                                                                                                                                                                                                                                  | COGNITIVE PERFORMANCE                                                                                                                                                                                  | EMOTION, MOOD & PREFERENCE                                                                                                                                                                                                                             |
|---------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NATURE IN THE SPACE | Visual Connection with Nature     | <ul style="list-style-type: none"> <li>Lowered blood pressure and heart rate (Brown, Barton &amp; Gladwell, 2013; van den Berg, Hartig, &amp; Staats, 2007; Tsunetsugu &amp; Miyazaki, 2005)</li> </ul>                                                           | Improved mental engagement/ attentiveness (Biederman & Vessel, 2006)                                                                                                                                   | Positively impacted attitude and overall happiness (Barton & Pretty, 2010)                                                                                                                                                                             |
|                     | Non-Visual Connection with Nature | <ul style="list-style-type: none"> <li>Reduced systolic blood pressure and stress hormones (Park, Tsunetsugu, Kasetani et al., 2009; Hartig, Evans, Jamner et al., 2003; Orsega-Smith, Mowen, Payne et al., 2004; Ulrich, Simons, Losito et al., 1991)</li> </ul> | Positively impacted cognitive performance (Mehta, Zhu & Cheema, 2012; Ljungberg, Neely, & Lundström, 2004)                                                                                             | Perceived improvements in mental health and tranquility (Li, Kobayashi, Inagaki et al., 2012; Jahncke, et al., 2011; Tsunetsugu, Park, & Miyazaki, 2010; Kim, Ren, & Fielding, 2007; Stigsdotter & Grahn, 2003)                                        |
|                     | Non-Rhythmic Sensory Stimuli      | <ul style="list-style-type: none"> <li>Positively impacted heart rate, systolic blood pressure and sympathetic nervous system activity (Li, 2009; Park et al, 2008; Kahn et al., 2008; Beauchamp, et al., 2003; Ulrich et al., 1991)</li> </ul>                   | Observed and quantified behavioral measures of attention and exploration (Windhager et al., 2011)                                                                                                      |                                                                                                                                                                                                                                                        |
|                     | Thermal & Airflow Variability     | <ul style="list-style-type: none"> <li>Positively impacted comfort, well-being and productivity (Heerwagen, 2006; Tham &amp; Willem, 2005; Wigö, 2005)</li> </ul>                                                                                                 | Positively impacted concentration (Hartig et al., 2003; Hartig et al., 1991; R. Kaplan & Kaplan, 1989)                                                                                                 | Improved perception of temporal and spatial pleasure (alliesthesia) (Parkinson, de Dear & Candido, 2012; Zhang, Arens, Huizenga & Han, 2010; Arens, Zhang & Huizenga, 2006; Zhang, 2003; de Dear & Brager, 2002; Heschong, 1979)                       |
|                     | Presence of Water                 | <ul style="list-style-type: none"> <li>Reduced stress, increased feelings of tranquility, lower heart rate and blood pressure (Alvarsson, Wiens, &amp; Nilsson, 2010; Pheasant, Fisher, Watts et al., 2010; Biederman &amp; Vessel, 2006)</li> </ul>              | Improved concentration and memory restoration (Alvarsson et al., 2010; Biederman & Vessel, 2006)<br>Enhanced perception and psychological responsiveness (Alvarsson et al., 2010; Hunter et al., 2010) | Observed preferences and positive emotional responses (Windhager, 2011; Barton & Pretty, 2010; White, Smith, Humphries et al., 2010; Karmanov & Hamel, 2008; Biederman & Vessel, 2006; Heerwagen & Orians, 1993; Ruso & Atzwanger, 2003; Ulrich, 1983) |
|                     | Dynamic & Diffuse Light           | <ul style="list-style-type: none"> <li>Positively impacted circadian system functioning (Figueiro, Brons, Plitnick et al., 2011; Beckett &amp; Roden, 2009)</li> <li>Increased visual comfort (Elyezadi, 2012; Kim &amp; Kim, 2007)</li> </ul>                    |                                                                                                                                                                                                        |                                                                                                                                                                                                                                                        |
|                     | Connection with Natural Systems   |                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                        | Enhanced positive health responses; Shifted perception of environment (Kellert et al., 2008)                                                                                                                                                           |
| NATURAL ANALOGUES   | Biomorphic Forms & Patterns       | *                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        | Observed view preference (Vessel, 2012; Joye, 2007)                                                                                                                                                                                                    |
|                     | Material Connection with Nature   |                                                                                                                                                                                                                                                                   | Decreased diastolic blood pressure (Tsunetsugu, Miyazaki & Sato, 2007)<br>Improved creative performance (Lichtenfeld et al., 2012)                                                                     | Improved comfort (Tsunetsugu, Miyazaki & Sato 2007)                                                                                                                                                                                                    |
|                     | Complexity & Order                | <ul style="list-style-type: none"> <li>Positively impacted perceptual and physiological stress responses (Salingaros, 2012; Joye, 2007; Taylor, 2006; S. Kaplan, 1988)</li> </ul>                                                                                 |                                                                                                                                                                                                        | Observed view preference (Salingaros, 2012; Hägerhäll, Laike, Taylor et al., 2008; Hägerhäll, Purcella, & Taylor, 2004; Taylor, 2006)                                                                                                                  |
| NATURE OF THE SPACE | Prospect                          | <ul style="list-style-type: none"> <li>Reduced stress (Grahn &amp; Stigsdotter, 2010)</li> </ul>                                                                                                                                                                  | Reduced boredom, irritation, fatigue (Clearwater & Coss, 1991)                                                                                                                                         | Improved comfort and perceived safety (Herzog & Bryce, 2007; Wang & Taylor, 2006; Petherick, 2000)                                                                                                                                                     |
|                     | Refuge                            | <ul style="list-style-type: none"> <li></li> <li></li> <li></li> </ul>                                                                                                                                                                                            | Improved concentration, attention and perception of safety (Grahn & Stigsdotter, 2010; Wang & Taylor, 2006; Petherick, 2000; Ulrich et al., 1993)                                                      |                                                                                                                                                                                                                                                        |
|                     | Mystery                           | <ul style="list-style-type: none"> <li></li> <li></li> </ul>                                                                                                                                                                                                      |                                                                                                                                                                                                        | Induced strong pleasure response (Biederman, 2011; Salimpoor, Benovoy, Larcher et al., 2011; Ikemi, 2005; Blood & Zatorre, 2001)                                                                                                                       |
|                     | Risk/Peril                        | *                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                        | Resulted in strong dopamine or pleasure responses (Kohno et al., 2013; Wang & Tsien, 2011; Zald et al., 2008)                                                                                                                                          |

Why?

1

Design to Minimise  
Waste

2

Design for  
Cyclability

3

Design to Reduce  
Chemical Impacts

4

Design to Reduce  
Energy and Water Use

5

Design that Explores  
Clean/Better  
Technologies

6

Design that Takes  
Models from  
Nature & History

7

Design for Ethical  
Production

8

Design to Reduce  
the Need to Consume

9

Design to Dematerialise  
and Develop  
Systems & Services

10

Design  
Activism

TED 10

<https://www.youtube.com/watch?v=e7vdKn0FpZ4>



<https://asknature.org/>

<https://asknature.org/resource/biomimicry-in-action-case-studies/>

# Activity

Ask  
Nature  
.org

## Activity

Use biomimicry, biophilic design and /or nature-inspired design to propose a redesign of something in your field

[https://www.learnbiomimicry.com/blog/biomimicry-examples-2025?srsId=AfmBOooClw8zBNOg2jDwcNcBlpMOiFNYtgmOL8fTIQF\\_VRvjLvULdu50](https://www.learnbiomimicry.com/blog/biomimicry-examples-2025?srsId=AfmBOooClw8zBNOg2jDwcNcBlpMOiFNYtgmOL8fTIQF_VRvjLvULdu50)



Share

<https://silphiumdesign.com/cases-successful-biomimicry-digital-design/>



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